

# VITALIS

An adaptive emergency triage agent. Android & iOS, React Native on Expo. This document is the complete feature inventory for whoever designs the interface.

|         |                  |             |
|---------|------------------|-------------|
| 9       | 6                | 161         |
| SCREENS | ROUTING OUTCOMES | TESTS GREEN |

## 00 What this product actually is

A patient opens the app during a medical emergency. The agent interviews them, and every question it asks depends on what they have already said. It calls a deterministic clinical engine to score risk, keeps a structured case record, decides among six fixed routing outcomes, and re-plans when the patient contradicts themselves.

**The guardrail, and it must be visible in the UI:** this is decision support in a simulated environment. The system never diagnoses and never prescribes. No screen may imply otherwise — not in a headline, not in a button label, not in an empty state.

It is not a chatbot with a medical prompt. The difference a judge is looking for is that this thing *holds state, changes its plan, and refuses to guess*. The interface has to make all three of those visible.

## 01 The one idea the interface must carry

Two things are tracked separately, and conflating them visually would break the product.

|                                                                                                                                                   |                                                                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Risk</b><br>Axis 1<br>How bad is this? Produced only by the clinical scoring engine. Green → Yellow → Red, changing live during the interview. | <b>Confidence</b><br>Axis 2<br>How much should the agent believe what it is being told? Driven by contradictions and evidence quality. Independent of risk. |
|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Why they must look different:** a low-confidence Green is not a green light. It is a reason to ask harder questions. If the UI renders confidence as just another severity color, that

distinction dies. Give them different visual channels entirely — risk owns the color language below; confidence needs its own form (a meter, a texture, a stated sentence).

|                                                                                                   |                                                                                                               |                                                                                                   |
|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| <div>GREEN • LOW RISK</div> <div>Self-care or routine consultation. Calm, unhurried layout.</div> | <div>YELLOW • ELEVATED</div> <div>Consultation within 24h, or urgent care now. Attention without alarm.</div> | <div>RED • CRITICAL</div> <div>Emergency. Full-bleed treatment, largest type, fewest words.</div> |
|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|

**Color is never the only signal.** Every tier is also labelled in words on screen. Red/green is the most common form of color blindness, and “which one was the bad one?” is not an acceptable question here.

02

Screen inventory

Nine screens. Four are built and working; five exist as rough layouts with correct information hierarchy and no visual design. The emergency path is strictly linear: Home → Emergency → Tracking.

|                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>Home</div> <div>BUILT</div> <div>Resting state. The app spends 99% of its life here.</div> <div><ul style="list-style-type: none"><li>• <b>Vitals dashboard</b> — BP, glucose, SpO<sub>2</sub>, heart rate</li><li>• <b>Medication reminders</b></li><li>• <b>Emergency button</b> — always reachable, dominates</li><li>• Quiet links to First aid, Emergency card, Language</li></ul></div> | <div>Emergency</div> <div>BUILT</div> <div>“Are you in an emergency?” → the adaptive interview.</div> <div><ul style="list-style-type: none"><li>• <b>11 quick-select symptom tags</b> (chest pain, coughing blood, paralysis, severe bleeding, head injury, burns, dizziness, shaking, vomiting, breathlessness, abdominal pain)</li><li>• <b>Live risk badge</b> that changes mid-interview</li><li>• <b>Agent’s question + why it asked</b></li><li>• <b>Press-and-hold dial</b>, 3 seconds</li><li>• Confidence alert state; degradation banner</li></ul></div> |
| <div>Tracking</div> <div>BUILT</div> <div>“Help is on the way.” Post-dispatch.</div> <div><ul style="list-style-type: none"><li>• <b>Map</b> with matched hospital marker</li><li>• <b>ETA, unit, status, contact number</b></li><li>• <b>Cancel Alert</b> — permanently visible, never scrolls away</li><li>• Live timeline</li></ul></div>                                                       | <div>First aid</div> <div>BUILT</div> <div>Works with the radio off. Bundled into the app.</div> <div><ul style="list-style-type: none"><li>• CPR, choking, bleeding, burns</li><li>• <b>Numbered steps</b>, one action per step</li><li>• “Call 112 first” where it applies</li></ul></div>                                                                                                                                                                                                                                                                        |

**Doctor Handoff**

ROUGH

A clinician reads this in seconds.

**Highest-value screen to design well.**

- **Severity /10, elapsed minutes, confidence** — lead with these
- **“Classic presentation” flag**
- **Patient’s own words, verbatim** — never paraphrased
- Allergies & medications; unresolved contradictions
- ICD-11 code; timeline; scoring-source disclosure

**Companion**

ROUGH

Proof the agent did not stop after triage.

- **Reassessment count + next check** — largest elements
- **Trends:** breathing, consciousness, bleeding — worsening / stable / improving
- Matched first-aid prompt
- Who is receiving live location

**Emergency card**

ROUGH

Opened by a paramedic on someone else’s phone.

- **Blood group & allergies** at the top, largest size
- Medications (RxNorm-normalised), chronic conditions
- Emergency contacts, tap to call
- Notes for responders; organ donor status

**Language**

ROUGH

English, Hindi, Telugu, Tamil.

- **Each option written in its own script** — someone who reads only Telugu cannot find “Telugu” in a list of English words
- Changing language never restarts a case

**Injury photo**

ROUGH

Optional extra observation, not a shortcut past triage.

- Capture / choose, then confirm
- **Explicit unavailable state** — no vision model on the account yet
- Explains what happens to the photo *before* it is taken

**Interface rules that cannot be designed away**

These are enforced in code as well as in design. A layout that breaks one of them will fail review regardless of how it looks.

**Press and hold for 3 seconds before any emergency call**

Never auto-dial on a single tap. This is a child-safety safeguard against pocket taps. Releasing early resets to zero — visibly. The duration is validated on the server too, so a UI that “helpfully” shortens it simply gets rejected.

**Cancel Alert stays visible at all times during tracking**

Not behind a menu, not below the fold. A cancel control you have to hunt for during an emergency is not available. It is a plain tap, deliberately — the gate exists to prevent accidental dispatch, and making it hard to stand down would be backwards.

**Degradation is stated, never hidden**

When a service is unavailable the app says so in a specific sentence written in advance, and shows what it used instead. There must be a banner treatment for this on every screen that can show a risk level. “Silently showing a worse answer” is the single worst failure this product can have.

**The agent shows its reasoning**

Every question comes with why it was asked. Every risk change comes with what triggered it. There is a live tool-call ledger — real external calls, timestamped — that should be visible, because it is proof rather than a claim.

**Escalation to a human is a real outcome, not an error state**

When the agent cannot resolve a contradiction on a high-risk case, it hands off rather than guessing. Design this as a deliberate, confident outcome. It must not look like a crash or a dead end.

**Tone adapts; rigor does not**

A panicked user gets shorter sentences and plainer words. They get the *identical* risk tier, allowed outcomes and safety gates as a calm one. Nothing in the UI may soften a gate because someone sounds distressed.

## Feature inventory

Everything the product does, and its real state today. **LIVE** works end to end. **PARTIAL** is built but standing in. **GAP** is not wired.

|                                                                                                                                                               |     |         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------|
| <b>Adaptive interview</b>                                                                                                                                     | 5.1 | LIVE    |
| Each question depends on prior answers and current uncertainty. Never a fixed script.<br>Generated live by Groq.                                              |     |         |
| <b>Deterministic risk scoring</b>                                                                                                                             | 5.1 | PARTIAL |
| Green/Yellow/Red, recomputed every turn. A rule-based engine, never the language model — that separation is the strongest claim this project makes.           |     |         |
| <b>Confidence-based escalation</b>                                                                                                                            | 5.1 | LIVE    |
| Contradictions raise a Confidence Alert and force re-questioning before any action.<br>Resolution requires an independent third source, not just a repeat.    |     |         |
| <b>Press-and-hold safety gate</b>                                                                                                                             | 5.2 | LIVE    |
| 3 seconds, validated on the server against a single shared constant.                                                                                          |     |         |
| <b>Quick-select symptom tags</b>                                                                                                                              | 5.2 | LIVE    |
| Eleven tags for someone who cannot type a sentence right now.                                                                                                 |     |         |
| <b>Simulated dispatch &amp; live tracking</b>                                                                                                                 | 5.2 | PARTIAL |
| Map, ETA, contact number, always-available cancel.                                                                                                            |     |         |
| <b>Hospital matching</b>                                                                                                                                      | 5.3 | GAP     |
| Nearest hospital by required specialty. Real coordinates from OpenStreetMap; specialty and bed availability are simulated and labelled as such on screen.     |     |         |
| <b>Doctor Handoff Timeline</b>                                                                                                                                | 5.3 | PARTIAL |
| Structured clinical card, not a chat transcript. Severity traces to the scoring engine by construction — no code path can inject an invented number.          |     |         |
| <b>Pre-arrival summary push</b>                                                                                                                               | 5.3 | GAP     |
| Sent to the matched hospital’s simulated endpoint.                                                                                                            |     |         |
| <b>Companion Mode</b>                                                                                                                                         | 5.4 | GAP     |
| Reassesses every 2–5 minutes after triage. Tracks breathing, consciousness and bleeding as direction of travel. Logic exists; nothing runs it on a timer yet. |     |         |

|                                                                                                                                                               |     |         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------|
| <b>Family Relay Mode</b>                                                                                                                                      | 5.5 | GAP     |
| If the patient stops responding, a caregiver takes over answering. Tone shifts toward clinical precision, because they are relaying rather than experiencing. |     |         |
| <b>Emergency contact alerts</b>                                                                                                                               | 5.5 | GAP     |
| WhatsApp and SMS via Twilio, with live location sharing. Credentials held; adapter not written.                                                               |     |         |
| <b>One-Tap Emergency Card</b>                                                                                                                                 | 5.6 | PARTIAL |
| Blood group, allergies, medications, contacts. Local — a card that needs a signal is not an emergency card.                                                   |     |         |
| <b>Multi-language</b>                                                                                                                                         | 5.6 | PARTIAL |
| English, Hindi, Telugu, Tamil. Translation preserves tone, not only literal meaning. Engine works; no picker wired.                                           |     |         |
| <b>Injury photo assessment</b>                                                                                                                                | 5.7 | GAP     |
| Describes what is visible — bleeding, swelling, burns. Never names a condition. Blocked: no vision model on the account.                                      |     |         |
| <b>Offline first aid</b>                                                                                                                                      | 5.8 | LIVE    |
| Bundled into the app, mirrored to device storage. Deliberately separate from the live agent loop.                                                             |     |         |
| <b>Plain-language explanations with citations</b>                                                                                                             |     | LIVE    |
| MedlinePlus (NIH) primary, Wikipedia as an explicitly-labelled fallback. Every claim carries its source.                                                      |     |         |
| <b>Medication name normalisation</b>                                                                                                                          |     | LIVE    |
| RxNorm/RxNav. Names only — interaction checking is <i>not</i> available and must never be implied in UI copy.                                                 |     |         |
| <b>Standardised diagnostic coding</b>                                                                                                                         |     | LIVE    |
| WHO ICD-11. Labels a category the scoring engine already derived; it does not diagnose.                                                                       |     |         |
| <b>Live case state across devices</b>                                                                                                                         |     | LIVE    |
| The phone observes Firestore directly. The server writes a new risk tier and the screen updates unprompted, with no refresh.                                  |     |         |

Tool-call ledger

LIVE

Every external call recorded with timing and outcome. Designed to be shown, as proof that tools are genuinely being called.

05

Six routing outcomes

A closed set. The agent never invents an outcome, and each carries its own confirmation gate. Every one of these needs a designed state.

| OUTCOME                  | GATE BEFORE IT HAPPENS            | TIER           |
|--------------------------|-----------------------------------|----------------|
| Self-care guidance       | None                              | Green          |
| Consultation within 24h  | None                              | Green / Yellow |
| Urgent care now          | Explicit tap to confirm           | Yellow         |
| Go to ER, self-transport | Explicit tap to confirm           | Yellow / Red   |
| Ambulance dispatch       | Press and hold, 3 seconds         | Red            |
| Escalate to a human      | None — the agent refuses to guess | Any            |

06

States every screen needs

The happy path is the easy part. These are where an emergency app is actually judged, and they are frequently the ones left undesigned.

| STATE            | WHEN IT APPEARS                                           | WHAT IT MUST NOT DO                                                       |
|------------------|-----------------------------------------------------------|---------------------------------------------------------------------------|
| Degraded         | A clinical service is unavailable and a fallback was used | Look like a minor warning. It changes how much the number can be trusted. |
| Confidence alert | The patient contradicted themselves                       | Read as an accusation. The person may be frightened, not lying.           |
| Escalated        | Agent handed off to a human rather than guess             | Look like an error or a dead end.                                         |
| Offline          | No connection                                             | Hide first aid. That is precisely when it matters.                        |

| STATE                | WHEN IT APPEARS                    | WHAT IT MUST NOT DO                                                       |
|----------------------|------------------------------------|---------------------------------------------------------------------------|
| Server unreachable   | Orchestrator down or wrong address | Say “something went wrong”. Say what to do.                               |
| Cancelled            | User stood down an alert           | Guilt-trip. Cancelling must feel safe, or people hesitate to call at all. |
| Waiting on the agent | A turn is being processed          | Freeze the cancel control.                                                |

07

## Current design tokens

What is in the code today. Treat this as a starting point to replace, not a constraint — except the tier colors, which carry meaning and are referenced throughout the codebase.

Green · low risk  
#15803D

Yellow · elevated  
#D97706

Red · critical  
#DC2626

Primary action  
#1D4ED8

Text  
#0F172A

Ground  
#F8FAFC

Spacing scale 4 / 8 / 12 / 16 / 24 / 32    Radius 8 / 12 / 16 / pill    Type 28 / 20 / 16 / 15 / 13 / 11 . System font throughout — no custom face is loaded yet, and one that supports Devanagari, Telugu and Tamil would need checking before adoption.

08

## Copy rules

- **Every message in an active emergency ends with one concrete next step.** Never end on reassurance alone.
- **No filler openings.** Not “I’m so sorry to hear that”. Say the thing.
- **Sentences shorten as urgency rises,** never lengthen.
- **Never claim a capability the system lacks** — particularly drug interaction checking, which is unavailable.

- **The patient’s own words appear verbatim** on the handoff card, in quotation marks, never cleaned up.
- **Unsourced claims are marked unsourced.** Explanations carry a visible source name.

09

Technical constraints for the designer

| CONSTRAINT                              | WHAT IT MEANS FOR DESIGN                                                                                        |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| React Native on Expo Go                 | No native modules. Effects like blur, complex shadows and custom fonts need checking before they are specified. |
| Phone-first, one hand, possibly shaking | Large targets, generous spacing, nothing critical near a screen edge.                                           |
| Free tier only                          | No paid icon sets, fonts or illustration licences.                                                              |
| Four scripts                            | Latin, Devanagari, Telugu, Tamil. Telugu and Tamil need more line height than Latin; tight leading will clip.   |
| Screens render in bright outdoor light  | Contrast budget is tighter than a normal app. Low-contrast grey-on-grey will disappear.                         |
| Map may fail to render                  | The tracking screen degrades to a coordinate card. That fallback needs a design too.                            |

**Deliverables that would help most, in order:** Doctor Handoff Timeline → Emergency screen with all its states → Home → Tracking → the remaining five. The handoff card is directly judged and all of its data already exists.